

Solutions & Notes of Homework 4

Junda Wu, Student ID: 210320621→25S065010

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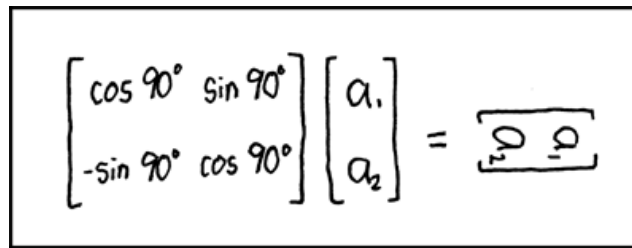

$$\begin{bmatrix} \cos 90^\circ & \sin 90^\circ \\ -\sin 90^\circ & \cos 90^\circ \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Figure 1: Picture from <https://matthbeck.github.io/325.html>, originally from <https://xkcd.com/>.

Problem 1

Show that the function that takes $((x_1, x_2), (y_1, y_2)) \in \mathbb{R}^2 \times \mathbb{R}^2$ to $|x_1 y_1| + |x_2 y_2|$ is not an inner product on $\mathbb{R}^2$.

✓ **Proof:** Denote the function as $(x_1, x_2) \circ (y_1, y_2)$. Let $v_1 = (1, 2), v_2 = (-1, -2), v_3 = (1, 1)$. Then

$$\left. \begin{aligned} v_1 \circ v_3 + v_2 \circ v_3 &= 3 + 3 = 6, \\ (v_1 + v_2) \circ v_3 &= (0, 0) \circ (1, 1) = 0, \end{aligned} \right\} \implies (v_1 + v_2) \circ v_3 \neq v_1 \circ v_3 + v_2 \circ v_3.$$

Therefore, the additivity in the first slot doesn't hold for the function, and this function is not an inner product on $\mathbb{R}^2$. ■

Problem 2

Suppose V is a real inner product space, show that:

- a) the inner product $\langle u + v, u - v \rangle = \|u\|^2 - \|v\|^2$ for every $u, v \in V$.
- b) if $u, v \in V$ have the same norm, then $u + v$ is orthogonal to $u - v$.
- c) use part(b) to show that the diagonals of a rhombus are perpendicular to each other.

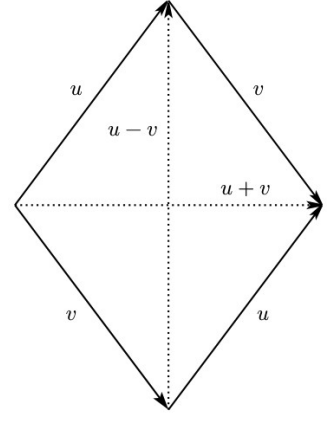
! **Note 2.1:** The inner product on a **real** inner product space satisfies additivity in both slots (note that the conjugate of a real number remains unchanged):

$$\langle u, v + w \rangle = \overline{\langle v + w, u \rangle} = \langle v + w, u \rangle = \langle v, u \rangle + \langle w, u \rangle = \overline{\langle u, v \rangle} + \overline{\langle u, w \rangle} = \langle u, v \rangle + \langle u, w \rangle.$$

✓ **Proof:** a) $\langle u + v, u - v \rangle = \langle u, u \rangle - \langle u, v \rangle + \langle v, u \rangle - \langle v, v \rangle = \langle u, u \rangle - \langle u, v \rangle + \langle u, v \rangle - \langle v, v \rangle = \|u\|^2 - \|v\|^2$.

b) This is immediate from part (a).

c) Suppose two adjacent sides of a rhombus are represented by two vectors $u, v \in \mathbb{R}^2$. Then u and v have the same norm. The two diagonals of the rhombus can be written as $u + v$ and $u - v$ respectively. From part (b), we know that $u + v$ is orthogonal to $u - v$, so the diagonals of a rhombus are perpendicular to each other. ■



Problem 3

Suppose $u, v \in V$, prove that the inner product $\langle u, v \rangle = 0$ if and only if $\|u\| \leq \|u + av\|$ for all $a \in \mathbb{F}$.

✓ **Proof:** If v or u equals 0, then it's easy to draw the conclusion. The proof below assume that $v \neq 0, u \neq 0$. First we assume that $\langle u, v \rangle = 0$. Then $\langle u, av \rangle = \bar{a}\langle u, v \rangle = 0$, hence u and av are orthogonal. Then from Pythagorean theorem, we have

$$\|u + av\|^2 = \|u\|^2 + \|av\|^2 = \|u\|^2 + |a|^2 \|v\|^2 \geq \|u\|^2.$$

Method 1: Conversely, if $\|u\| \leq \|u + av\|$ for all $a \in \mathbb{F}$, this implies

$$\|u + av\|^2 - \|u\|^2 = |a|^2 \|v\|^2 + a\langle v, u \rangle + \bar{a}\langle u, v \rangle \geq 0.$$

If $v = 0$, then $\langle u, v \rangle = 0$. If $v \neq 0$, plug $a = -\langle u, v \rangle / \|v\|^2$ into the previous equation, we obtain

$$-\frac{|\langle u, v \rangle|^2}{\|v\|^2} \geq 0.$$

Hence $\langle u, v \rangle = 0$.

Method 2: Conversely, assume that $\|u\| \leq \|u + av\|$. We write the orthogonal decomposition of u from 6.13:

$$u = cv + w, \langle w, v \rangle = 0, c \in \mathbb{F}.$$

Then we have

$$\|cv + w\| \leq \|cv + w + av\|.$$

From the arbitrariness of a , take $a = -c$ and we have

$$\|cv + w\| \leq \|w\|. \quad (1)$$

Since $\langle cv, w \rangle = c\langle v, w \rangle = 0$, cv and w are orthogonal. Then from Pythagorean theorem, we have

$$\|cv + w\| = \|cv\| + \|w\|. \quad (2)$$

It follows from (1) and (2) that

$$\|cv\| \leq 0 \implies \|cv\| = 0 \implies c = 0 \implies u = w.$$

Thus $\langle u, v \rangle = 0$, completing the proof. ■

! **Note 3.1:** Draw a picture in $\mathbb{R}^2$ may help. If u has nonzero “component” along v , then an appropriate selection of a can “cancel” this component and leads to a smaller norm.

Problem 4

Suppose $u, v \in V$, prove that $\|au + bv\| = \|bu + av\|$ for all $a, b \in \mathbb{R}$ if and only if $\|u\| = \|v\|$.

✓ **Proof:** Note that $\|au + bv\| = \|bu + av\| \iff \langle au + bv, au + bv \rangle = \langle bu + av, bu + av \rangle$. We have

$$\begin{aligned} \langle au + bv, au + bv \rangle &= a \langle u, au + bv \rangle + b \langle v, au + bv \rangle \\ &= a \overline{\langle au + bv, u \rangle} + b \overline{\langle au + bv, v \rangle} \\ &= \overline{aa \langle u, u \rangle} + \overline{b \langle v, u \rangle} + \overline{ba \langle u, v \rangle} + \overline{b \langle v, v \rangle} \\ &= a^2 \langle u, u \rangle + ab \langle u, v \rangle + b^2 \langle v, v \rangle + ab \langle v, u \rangle. \end{aligned} \quad (3)$$

Similarly we have

$$\langle bu + av, bu + av \rangle = b^2 \langle u, u \rangle + ab \langle u, v \rangle + a^2 \langle v, v \rangle + ab \langle v, u \rangle. \quad (4)$$

It follows from (3) and (4) that

$$\langle au + bv, au + bv \rangle = \langle bu + av, bu + av \rangle \iff (a^2 - b^2)(\langle u, u \rangle - \langle v, v \rangle) = 0.$$

Given this, it will suffice to show that $(a^2 - b^2)(\langle u, u \rangle - \langle v, v \rangle) = 0$ for all $a, b \in \mathbb{R}$ if and only if $\|u\| = \|v\|$.

The reverse implication is clear; for the forward implication, simply assign $a = 1, b = 0$. ■

Problem 5

Suppose $u, v \in V$, $\|u\| = \|v\| = 1$ and $\langle u, v \rangle = 1$, prove that $u = v$.

✓ **Proof (1):** $|\langle u, v \rangle| = 1 = \|u\| \|v\|$, then from Cauchy-Schwarz inequality, u must be a scalar multiple of v , i.e., $u = av, a \in \mathbb{F}$. Then $\langle u, v \rangle = \langle av, v \rangle = a \langle v, v \rangle = a = 1$. So $u = v$, as desired. ■

✓ **Proof (2):** Observe that $\langle u - v, u - v \rangle = \langle u, u \rangle - \langle u, v \rangle - \langle v, u \rangle + \langle v, v \rangle = \|u\|^2 - 2 \operatorname{Re} \langle u, v \rangle + \|v\|^2 = 0$. It follows from definiteness that $u - v = 0$. ■

Problem 6

Find vectors $u, v \in \mathbb{R}^2$ such that u is a scalar multiple of $(1, 3)$, v is orthogonal to $(1, 3)$, and $(1, 2) = u + v$.

✓ **Solution:** Let $u = k(1, 3) = (k, 3k) (k \in \mathbb{R})$, then $v = (1 - k, 2 - 3k)$. Because v is orthogonal to $(1, 3)$, then $\langle v, (1, 3) \rangle = 0$, so $1 - k + 6 - 9k = 0, k = \frac{7}{10}$. So $u = (\frac{7}{10}, \frac{21}{10}), v = (\frac{3}{10}, -\frac{1}{10})$.

! **Note 6.1:** Please refer to 6.13 of *Linear Algebra Done Right* (fourth edition).

Problem 7

Prove that $(x_1 + \cdots + x_n)^2 \leq n(x_1^2 + \cdots + x_n^2)$ for all positive integers n and all real numbers $x_1, \dots, x_n$.

✓ **Proof:** Let $u = (x_1, x_2, \dots, x_n), v = (1, 1, \dots, 1)$. Then from Cauchy-Schwarz inequality, we find

$$|\langle u, v \rangle| = |x_1 + x_2 + \dots + x_n| \leq \|u\| \|v\| = \sqrt{n} \sqrt{x_1^2 + \dots + x_n^2}.$$

Take the square of both sides of the equation and we complete the proof. ■

Problem 8

Suppose V is a real inner product space, prove that

$$\langle u, v \rangle = \frac{\|u + v\|^2 - \|u - v\|^2}{4}$$

for all $u, v \in V$.

✓ **Proof:**

$$\begin{aligned} \|u + v\|^2 - \|u - v\|^2 &= \langle u + v, u + v \rangle - \langle u - v, u - v \rangle \\ &= (\langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle) - (\langle u, u \rangle - \langle u, v \rangle - \langle v, u \rangle + \langle v, v \rangle) \\ &= (\langle u, u \rangle + 2\langle u, v \rangle + \langle v, v \rangle) - (\langle u, u \rangle - 2\langle u, v \rangle + \langle v, v \rangle) \\ &= 4\langle u, v \rangle. \end{aligned}$$

! **Note 8.1:** The identity is called **polarization identity**, which allows one to reconstruct the inner product from the norm.

! **Note 8.2:** Suppose that V is a complex inner product space. Then for all $u, v \in V$,

$$\langle u, v \rangle = \frac{\|u + v\|^2 - \|u - v\|^2 + \|u + \mathrm{i}v\|^2 - \|u - \mathrm{i}v\|^2}{4}.$$

Try to prove it by yourself.